PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	A systems approach to health service design, delivery and improvement: A systematic review and meta-analysis
AUTHORS	Komashie, Alexander; Ward, James; Bashford, Tom; Dickerson, Terry; Kaya, Gulsum; Liu, Yuanyuan; Kuhn, Isla; Gunay, Aslı; Kohler, Katharina; Boddy, Nicholas; O'Kelly, Eugenia; Masters, Joseph; Dean, John; Meads, Catherine; Clarkson, John

VERSION 1 – REVIEW

REVIEWER	DonHee Lee South Korea
REVIEW RETURNED	22-Apr-2020

GENERAL COMMENTS	Review of “A systems approach to health service design, delivery, and improvement: A systematic review and meta-analysis” I appreciate the work that went into this paper. This study is evaluated as a study that analyzes the usefulness of accessing the system without limitation of language, participant types, outcome types, or any particular healthcare domain for improving healthcare service through tremendous previous research. However, there are a number of elements that might strengthen the manuscript particularly about the research model part. The focus of the paper is currently problematic. The literature review also needs tightening. You do a great job making sure you define concepts as you introduce them, but typically the operationalization of the constructs would not appear so early in the paper. Additional clarity and specific support could be provided for why the specific measures are used. It is not clear how the current paper contributes to something new and important for healthcare improvement. In addition, I am not sure what the results suggested for healthcare design, delivery, and improvement in the manuscript. Discussion of results is limited, furthering the concern of minimal contribution to theory or practice. Again, I suggest careful development of carefully chosen relationships, each supported by theoretical logic, enabling richer discussion of the specific findings that can guide managers and other researchers. I hope the suggestions above help in that process and I wish the authors well in their efforts.

REVIEWER	Hiske Calsbeek Radboud university medical center, The Netherlands
REVIEW RETURNED	14-Jun-2020

GENERAL COMMENTS	Extensive, interesting and relevant review about an integrated approach to study the impact of a systems approach to healthcare improvement. Overall a very nice study but the writing is preliminary. The Abstract misses relevant information. Please consider the following questions, comments: OBJECTIVES: what is meant by 'evidence-base', what kind of outcomes? In what kind of healthcare settings? METHODS: please clarify what kind of data were extracted, what is meant by 'control data', to what extent do the 'patient and/or service outcomes' match the objective ('evidence-base')? Were sensitivity analyses performed? RESULTS section only mentions results of a meta-analysis of included before-after studies (with huge heterogeneity), what were the results of the other studies included? What was the quality of the studies included (Conclusions section reports 'poor' but this is new information)? What characterized the systems approaches that were studied? In what settings? The presented ORs in the Results section are not complete (3 results while 5 meta analyses). Other (also some minor) points: explain abbreviations the first time (e.g. CASP checklist p.6); first table in supp file 3 is not listed on p.5 (Table a on p.5 matches with Table 2 in supp file 3); tables on service outcomes (d and e on p.6) also contain information on 'research and cost outcomes' according to the title in supp file 3), table g on p.6 on quality assessment is not only based on CASP but also a checklist for RCTs. This table in de supp file should be Table 6 instead of Table 1 (and change the title of this Table as CASP was not the only checklist used). Explain why CASP was chosen (given the variation in study types included); Explain scoring codes used in sup file 3 (e.g. Table 5; what is meant by systems approach 0 or 1? And: systems intervention 0 or 1)? Provide some examples or a description of patient outcomes and service outcomes, e.g. why is neck/shoulder pain a service outcome in Table 4b supp file 3 see first study, and not a patient outcome? How to interpret meta-analyses results on these 'categories' of outcomes? Table 1 (and corresponding text) in the main text lacks information on most of the characteristics that are mentioned in the discussion section (in limitations paragraph, on heterogeneity), such as setting, participants, interventions. Also, an important part of the results and discussion sections is about 2 RCTs, Table 1 only mentions concurrent control as a study type, which studies are these RCTs? And which studies (in Table 1) are included in which meta-analysis? Text on results under Table 1 contains discussion-like text about how to interpret results. Results on outcomes are unclear, what outcomes? Qualitative synthesis is poor, what is meant by a significant number of success factors? Authors state that a full synthesis is beyond the scope of this review, but what is within the scope? What is the quality of the evidence (what is meant by 'a challenge')? The search in May 2019 needs an update before publication.

REVIEWER	Yasuaki SAIJO Asahikawa Medical University

REVIEW RETURNED	30-Jun-2020

GENERAL COMMENTS	I think that statistical methods in this paper are almost correct. However, in the funnel plot, only 16 of all 23 studies were included. To confirm publication bias, Begg's test etc. should be done. (Epidemiological point) Though odds ratios and standardized mean values of various outcomes can be aggregable, a meta-analysis should target one specific intervention (exposure) and one outcome. I think that relationships between various interventions and outcomes in various setting should not be analyzed in a meta-analysis.

REVIEWER	Yan Chen The University of Auckland, New Zealand
REVIEW RETURNED	01-Jul-2020

GENERAL COMMENTS	Thanks for the invitation to review this manuscript. My comments are mainly about the statistical methods and analyses used in the current investigation. The authors followed the recommended steps to conduct the meta-analysis and described in detail about the inclusion/exclusion criteria for data extraction. As I read through the paper, I kept wondering about whether there is a need to conduct a series of meta-analyses given the high heterogeneity of the data. Was the high heterogeneity pre-specified in the protocol registration? What are the causes of high heterogeneity? A strength of meta-analysis comes from its ability to combine effect sizes of a particular intervention from studies that use different designs and outcome measures. It would have been beneficial for the authors to justify their decision to run separate meta-analysis for studies with continuous outcome variables and those with categorical variables. According to the Cochrane Handbook (9.4.6), dichotomous outcomes can be combined with continuous outcomes to improve the validity and power of the meta-analysis. Did the authors try to contact authors about unpublished data or to request raw data to estimate effect sizes to combine different types of outcomes? How did the authors assign adifferent weighting to studies based on their CASP scores? Instead of running separate meta-analyses, I wonder if it is possible to combine these analyses into one and run meta-regressions to identify the impact of study design and different outcome measures on the meta-analysis results. The authors claim that "only studies that measured patient outcomes using concurrent controls showed an overall effect that slightly favoured the controls, as shown in Figure 3". However, Figure 6 also shows no impact of a systems approach on service and resource in studies with concurrent controls. Given that some studies have been included in multiple meta-analyses, I wonder whether it is possible for this approach to have inflated the meta-analysis results.

VERSION 1 – AUTHOR RESPONSE

	REVIEWER 1 COMMENTS	
1	I appreciate the work that went into this	Thank you.

	paper.	
2	This study is evaluated as a study that analyzes the usefulness of accessing the system without limitation of language, participant types, outcome types, or any particular healthcare domain for improving healthcare service through tremendous previous research. However, there are a number of elements that might strengthen the manuscript particularly about the research model part.	We thank the peer reviewer for their comments.

3	The focus of the paper is currently problematic.	We acknowledge that the systematic review question is wide, but we do not think it is a problem, given the caveats we have discussed.
4	The literature review also needs tightening.	We are unclear as to exactly what the peer reviewer is asking for here. Also, this is a systematic review rather than a literature review.
5	You do a great job making sure you define concepts as you introduce them, but typically the operationalization of the constructs would not appear so early in the paper.	This systematic review is the first of its kind and the concept of a systems approach is not often well-defined in the literature, so we felt that it was important to spell out the systems constructs from the start.
	Additional clarity and specific support	In a systematic review we can only report on the specific measures

6	could be provided for why the specific measures are used.	used in the included primary studies.
7	It is not clear how the current paper contributes to something new and important for healthcare improvement.	We discuss in the discussion section the current literature in this new area and show how our systematic review takes the evidence further by conducting meta-analyses. Please see page 13 in the revised manuscript.
8	In addition, I am not sure what the results suggested for healthcare design, delivery, and improvement in the manuscript.	All we can show at this stage is that a systems approach seems to be beneficial, for both patient and service outcomes. We cannot be more specific around individual service improvement interventions in specific clinical areas because of insufficient evidence. Often a systematic review is a prompt to encourage further primary studies in key areas.
9	Discussion of results is limited, furthering the concern of minimal contribution to theory or practice.	In a systematic review it is usual to limit the discussion to the main findings of the systematic review and their general implications to practice. In this systematic review, as we mention above, one main contribution might well be to encourage RCTs in key areas.
10	Again, I suggest careful development of carefully chosen relationships, each supported by theoretical logic, enabling richer discussion of the specific findings	We are unclear as to the relationships mentioned by the peer reviewer.

	that can guide managers and other researchers.	
11	I hope the suggestions above help in that process and I wish the authors well in their efforts.	We thank the peer reviewer for their comments.
	REVIEWER 2 COMMENTS	
1	Extensive, interesting and relevant review about an integrated approach to study the impact of a systems approach to healthcare improvement.	We thank the peer reviewer for their encouraging comments.
2	Overall a very nice study but the writing is preliminary.	We have extensively edited the manuscript now and hope this version appears more polished.

	The Abstract misses relevant information. Please consider the following questions, comments: OBJECTIVES: what is meant by 'evidence-base', what kind of outcomes? In what kind of healthcare settings? METHODS: please clarify what kind of data were extracted,	We agree and this has now been edited to conform to PRISMA criteria. We think this refers to the abstract, which has now been edited. We are limited to word count in the abstract but we have ensured that all of these details are in the methods section. Evidence-base comes from Evidence-based health. An evidence base is "A body of information drawn from routine statistical analyses, published studies and "grey" literature - about what is already known about factors affecting health".

3	what is meant by 'control data', to what extent do the 'patient and/or service outcomes' match the objective ('evidence-base')? Were sensitivity analyses performed? RESULTS section only mentions results of a meta-analysis of included before-after studies (with huge heterogeneity), what were the results of the other studies included? What was the quality of the studies included (Conclusions section reports 'poor' but this is new information)? What characterized the systems approaches that were studied? In what settings? The presented ORs in the Results section are not complete (3 results while 5 meta analyses).	Outcomes had to be relevant to the study conducted, for example if the intervention was a safe patient handling and mobilisation programme the outcomes were various injury rates. Outcomes for each study can be seen in Tables 3a, 3b, 4a and 4b in the online supplementary file 3. You can see the data extracted in these and the other tables in online supplementary file 3. The word 'control has now been edited out of the abstract and methods section and exchanged for the word 'comparator' to be clearer. Please see for example third and last paragraphs on page 5. Study outcomes are part of the evidence base. Given the exploratory nature of the meta-analyses, no sensitivity analyses were performed. In the abstract results we now include a sentence on the general results and the study quality. The settings are various and have now been added to table 1 in the main text to aid the reader. We now only present 2 meta-analyses.

4	Other (also some minor) points: explain abbreviations the first time (e.g. CASP checklist p.6);	CASP stands for Critical Appraisal Skills Programme. This is now written in full on p6.
5	first table in supp file 3 is not listed on p.5 (Table a on p.5 matches with Table 2 in supp file 3);	The list of tables on page 6 has now been updated to correspond to the tables in supplementary file 3 and are in the correct order.
6	tables on service outcomes (d and e on p.6) also contain information on 'research and cost outcomes' according to the title in supp file 3),	Yes, tables on service outcomes include outcomes related to resource usage and cost impact of interventions where reported.
7	table g on p.6 on quality assessment is not only based on CASP but also a checklist for RCTs. This table in de supp file should be Table 6 instead of Table 1 (and change the title of this Table as CASP was not the only checklist used).	CASP provides checklists for different study designs. We used three of those – for cohort studies, case control studies and RCTs. These are now in tables numbered 6, 7 and 8 in supplementary file 3. We acknowledge that the list of tables on page 6 of the original manuscript was confusing as it was not consistent with the numbering in the supplementary file 3.This has now been updated and numbered in the right order in which they appear in the supplementary file. Please see page 6 of the revised manuscript.
	Explain why CASP was chosen (given the variation in study types included); Explain scoring codes used in sup file 3 (e.g. Table 5; what is meant by systems	CASP was chosen because it has a suite of checklists appropriate for different study designs. This point is now made in the methods text. After studies were identified for inclusion, the team assessed each study against our working definition of a systems approach as

8	approach 0 or 1? And: systems intervention 0 or 1)?	presented on page 4. The assessment used a binary scale (0 or 1) to indicate whether a study considered the concept of systems in the approach to the development of its intervention or in the implementation of an intervention, use of design and consideration of risk (please see first paragraph on page 6 of revised manuscript). Therefore, “Systems-Approach” scored as “0” means the study did not consider systems in the development of the reported intervention. A score of “1” means it did. A note has been added under table 5 to briefly explain these.
9	Provide some examples or a description of patient outcomes and service outcomes, e.g. why is neck/shoulder pain a service outcome in Table 4b supp file 3 see first study, and not a patient outcome?	Examples now given in text (please see paragraph after list of tables on page 6 of the revised manuscript). It is acknowledged that service and patient outcomes are similar, but in the example given, we judged it as a service outcome because the study was about improving moving and handling injuries, and these were worker injuries. (Dennerlein 2017).
10	How to interpret meta-analyses results on these ‘categories’ of outcomes?	We are much clearer now in the text about these being exploratory meta-analyses to show the direction of effect, rather than placing any importance in the numerical estimate of effect size. Please see, for example, first two paragraphs on page 13 of the revised manuscript.
	Table 1 (and corresponding text) in the main text lacks information on most of	Table 1 has now been updated with additional columns for “health setting” and “intervention”.

11	the characteristics that are mentioned in the discussion section (in limitations paragraph, on heterogeneity), such as setting, participants, interventions.	
12	Also, an important part of the results and discussion sections is about 2 RCTs, Table 1 only mentions concurrent control as a study type, which studies are these RCTs? And which studies (in Table 1) are included in which meta-analysis?	The two RCTs are Rothemich and Rustagi (See the CASP results table for RCTs in online supplementary file 3). This is now added to Table 1. You can tell which studies are included in the meta-analyses from the Forest plot figures 2 and 3.
13	Text on results under Table 1 contains discussion-like text about how to interpret results.	Thank you for pointing this out, it has now been edited.
14	Results on outcomes are unclear, what outcomes?	The outcomes are listed in online supplementary file 3. Because they are ALL different, it's not really possible to be anything other than rather general about the outcomes in the text.
15	Qualitative synthesis is poor, what is meant by a significant number of success factors?	References to "qualitative synthesis" have been removed and description of some observations from included studies has been improved, and hopefully is much clearer. "Success factors" referred to factors reported by studies with positive outcomes as contributing to success. This has now been made clearer. Please see page 7.
16	Authors state that a full synthesis is beyond the scope of this review, but	This sentence has now been deleted and section improved. See above.

	what is within the scope?	
17	What is the quality of the evidence (what is meant by 'a challenge')?	The quality of the evidence is assessed using the CASP questionnaires, and generally refers to whether a study's results can be believed, based on how that study was conducted. The challenge is around appropriate interpretation of results – this sentence has now been amended. Please see page 13 of the revised manuscript.
18	The search in May 2019 needs an update before publication.	We conducted an update search on 24th July 2020 and have found very little additional evidence, in particular nothing that would affect the meta-analysis direction and the conclusions of the systematic review. We have added this comment to the manuscript (see page 5 of the revised manuscript) but we have not formally updated the analyses and results in the text. We are happy to do this if the editor considers it sufficiently important, but it will delay publication.
	REVIEWER 3 COMMENTS	
1	I think that statistical methods in this paper are almost correct. However, in the funnel plot, only 16 of all 23 studies were included.	The funnel plot has been redone using the largest meta-analysis (service outcomes) and adding studies from the patient outcomes meta-analysis where there was no service outcome for that study. The result is 23 included studies in the funnel plot, and it looks very similar to the previous one so the text for this remains the same. Begg's test really is not a very accurate test for publication bias, see *an Enst, W.A., Ochodo, E., Scholten, R.J. *et al.* Investigation of

	To confirm publication bias, Begg's test etc. should be done.	publication bias in meta-analyses of diagnostic test accuracy: a meta-epidemiological study. *BMC Med Res Methodol* **14,** 70 (2014). https://doi.org/10.1186/1471-2288-14-70 The power is very low if there are small numbers of studies. Begg and Mazumdar (1994) say that their test is 'fairly powerful with 75
		studies, moderate power with 25 studies'. We only have 23 studies so less than moderate power.
2	(Epidemiological point) Though odds ratios and standardized mean values of various outcomes can be aggregable, a meta-analysis should target one specific intervention (exposure) and one outcome. I think that relationships between various interventions and outcomes in various setting should not be analyzed in a meta-analysis.	Ordinarily that would be the case, but as there were no interventions and outcomes in common, this is not possible, which is why it is clearly described as exploratory. Given the heterogeneity of the evidence base, at this stage the only way to indicate the general direction results of a systems approach is to present exploratory meta-analyses. When the evidence base improves, single interventions and outcomes can then be assessed individually. Please see page 13 of the revised manuscript.
	REVIEWER 4 COMMENTS	
	Thanks for the invitation to review this manuscript. My comments are mainly	We thank the peer reviewer for their comments. We have now limited ourselves to 2 meta-analyses only. The high statistical

1	about the statistical methods and analyses used in the current investigation. The authors followed the recommended steps to conduct the meta-analysis and described in detail about the inclusion/exclusion criteria for data extraction. As I read through the paper, I kept wondering about whether there is a need to conduct a series of meta-analyses given the high heterogeneity of the data. Was the high heterogeneity pre-specified in the protocol registration? What are the causes of high heterogeneity?	heterogeneity is caused by heterogeneity in patients, settings, interventions and outcome measures.
2	A strength of meta-analysis comes from its ability to combine effect sizes of a particular intervention from studies that use different designs and outcome measures. It would have been beneficial for the authors to justify their decision to run separate meta-analysis	We decided that exploratory meta-analyses only really needed to indicate the general direction of effect, the magnitude of effect size is not useful and gives an artificial precision that goes beyond the current evidence we have. As the continuous outcome meta-analysis for service outcomes has a similar direction of effect as the categorical outcomes for service outcomes, we decided just to present the categorical outcomes meta-analysis. We thought that nothing would be gained by adding continuous to categorical

	for studies with continuous outcome variables and those with categorical variables.	outcomes in one meta-analysis, particularly as only 3 studies in the continuous outcomes meta-analysis are not in the categorical outcomes meta-analysis. Please see the top half of page 13 in the revised manuscript.
3	According to the Cochrane Handbook (9.4.6), dichotomous outcomes can be combined with continuous outcomes to improve the validity and power of the meta-analysis.	Please see comment above.
4	Did the authors try to contact authors about unpublished data or to request raw data to estimate effect sizes to combine different types of outcomes?	No we didn't try to contact authors. As we had sufficient studies to demonstrate the general direction of effect, we did not think this was warranted.
5	How did the authors assign a different weighting to studies based on their CASP scores?	Quality assessment using CASP is intended to demonstrate a general appreciation of the quality of one study compared to another. Any reference to scores and weighting from scores has now been removed. Each study was quality assessed by two researchers, first individually and then resolving any differences.
	Instead of running separate meta-analyses, I wonder if it is possible to	This comment is interesting, particularly when compared to peer reviewer 3 who suggested we should not be running combined

6	combine these analyses into one and run meta-regressions to identify the impact of study design and different outcome measures on the meta-analysis results.	meta-analyses with different interventions and outcome measures. These two approaches highlight the general issues about when to combine and when not to combine in meta-analyses. As a result we have taken the decision to just present two exploratory meta-analyses and focus on the direction of effect rather than the

		numerical effect size. We think the heterogeneity of patients, interventions and outcomes means that a combined meta-analysis would give artificial precision. Meta-regression was not pre-specified and is relatively low powered, so requires sufficient data. Since ALL of the systematic review outcome measures are different, there would be insufficient power to demonstrate anything.
7	The authors claim that “only studies that measured patient outcomes using concurrent controls showed an overall effect that slightly favoured the controls, as shown in Figure 3”. However, Figure 6 also shows no impact of a systems approach on service and resource in studies with concurrent controls.	This meta-analysis only had 4 studies, and the overall effect size (CIs) was 1.29 [0.35, 4.80]. i.e. wide CIs and not statistically significant so we decided it was not helpful so have now omitted it. Similarly, with the meta-analysis with 3 included studies (0.66 [0.35, 1.24]).
	Given that some studies have been included in multiple meta-analyses, I	This is another reason why we have now limited ourselves to 2 meta-analyses.

8	wonder whether it is possible for this approach to have inflated the meta- analysis results.	

VERSION 2 – REVIEW

REVIEWER	Yan Chen The University of Auckland, New Zealand
REVIEW RETURNED	15-Oct-2020

GENERAL COMMENTS	The authors have addressed my concerns and I'm happy to recommend its publication.